

Mohammadreza Nemati

Santa Clara, CA 95054 | Mobile: 567-204-7520 | mnrnemati747@gmail.com | www.linkedin.com/in/mnemati

EXPERIENCE

Senior AI Engineer

05/2025 - Present

Volkswagen Group of America

Belmont, CA

- Built GUI-agent system for real-device app and browser automation on Android, combining screen understanding, element grounding, ADB action execution, action verification, and error recovery across different Android platforms.
- Developed long-horizon planning workflows for GUI agents, including dynamic task planning, task decomposition, app/browser navigation, and completion-evidence tracking for multi-step user goals.
- Fine-tuned multimodal foundation models using SFT, DPO, and GRPO for specialized planner, reasoner, grounder, verifier, and error recovery components in real-device GUI automation.
- Applied pruning, knowledge distillation, and quantization-aware training to compress VLMs for constrained deployment, then served custom checkpoints with vLLM continuous batching to improve throughput and latency for GUI-agent inference.

Artificial Intelligence Research Assistant

08/2022 - 06/2025

Case Western Reserve University

Cleveland, Ohio

- Optimized LLM inference memory by analyzing KV-cache key/value norm imbalance, achieving 2× memory reduction with >94% accuracy retention across models up to 70B parameters [\[paper link\]](#)
- Developed LIT-LVM using latent variable embeddings, structured regularization, and GNN-based higher-order interactions, outperforming GNN, Factorization Machine, and elastic net baselines in ranking and recommendation tasks [\[paper link\]](#)
- Developed DEASurv, a deep multimodal model leveraging entity-specific embeddings, attention mechanisms, and contrastive loss to improve prediction on large-scale medical datasets

Software Engineer - AI/ML Intern

02/2024 - 07/2024

Sherwin-Williams

Cleveland, Ohio

- Designed and deployed real-time vision-transformer inference pipeline for SAM, tuning latency and improving segmentation quality by 7% IoU through inference evaluation and pipeline optimization.
- Built multimodal system integrating LLaVA with vision transformers, implementing cross-modal alignment for natural-language visual grounding and object segmentation.

Data Science Research Intern

05/2023 - 09/2023

Beyond Limits AI

Glendale, California

- Developed and deployed Graph Neural Network models (GCN, GAT, GraphSAGE) using PyTorch Geometric for graph-structured data, achieving 10% accuracy improvement over baseline approaches
- Fine-tuned Llama-2 using PyTorch distributed training with FSDP/DDP for multi-GPU optimization and quantization for efficient deployment.

SKILLS

Programming: Python, C++, SQL

ML Engineering: PyTorch, Distributed Training (FSDP/DDP), Fine-tuning, Quantization, Knowledge Distillation, Computer Vision

Agentic Systems: GUI Agents, LangGraph, Multi-Agent Workflows, Long-Horizon Planning, Dynamic Planning, Task Decomposition, UI Grounding, Action Verification, Error Recovery

LLM/VLM Systems: vLLM, Prompting, Multimodal Models, Model Serving, Performance Evaluation

EDUCATION

Case Western Reserve University – PhD, Computer Science

Cleveland, OH | 2025

University of Toledo – MS, Computer Science

Toledo, OH | 2020

Shiraz University – BS, Electrical Engineering

Shiraz, Iran | 2017